

Homework 1. Markov chains

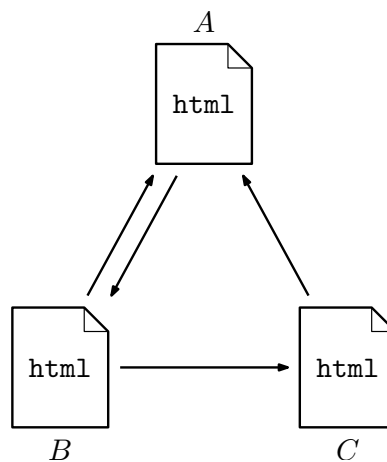


Figure 1: Web of linked documents. An arrow from a document x to a document y indicates that document x links to document y .

Consider the “web” of linked documents depicted in Fig. 1. In this representation, an arrow from a document x to a document y means that there is a link to document y in document x . In this homework, you will analyze the behavior of a bot navigating this web of documents. Upon arriving at a document, the bot will select, uniformly at random, one of the links in that document and follow that link.

Exercise 1.

- (a) Write down a Markov chain model that describes the motion of the bot described above in that web.
- (b) Let T_{xx} denote the average number of steps that the bot takes to return to document x when starting in x . Compute T_{AA} , T_{BB} and T_{CC} , indicating the relevant computations.

- (c) Note that, the longer it takes the bot to return to a document, the less time it spends in that particular document. In fact, the percentage of time that a bot spends in document x —let us denote it as μ_x —is inversely proportional to T_{xx} . Compute μ_A , μ_B and μ_C using the results from question (b). Show that the resulting distribution is invariant for the chain.

Solution:

- (a) The Markov chain is specified as a pair $(\mathcal{X}, \mathbf{P})$, where the states correspond to train positions. We have that

$$\mathcal{X} = \{A, B, C\},$$

where the states are labeled after the corresponding documents. The transition probabilities can be extracted from the provided description of the motion of the bot, yielding

$$\mathbf{P} = \begin{bmatrix} 0 & 1 & 0 \\ 0.5 & 0 & 0.5 \\ 1 & 0 & 0 \end{bmatrix}.$$

- (b) Denoting as T_{xy} the average number of steps that the bot takes to go from document x to document y , we have that

$$T_{AA} = 1 + T_{BA}, \quad T_{BA} = 0.5 \times 1 + 0.5 \times (1 + T_{CA}), \quad T_{CA} = 1,$$

yielding $T_{AA} = 2.5$. Similarly,

$$T_{BB} = 0.5 \times (T_{AB} + 1) + 0.5 \times (T_{CB} + 1), \quad T_{AB} = 1, \quad T_{CB} = 1 + T_{AB} = 2,$$

yielding $T_{BB} = 2.5$. Finally,

$$T_{CC} = 1 + T_{AC}, \quad T_{AC} = 1 + T_{BC} \quad T_{BC} = 0.5 \times 1 + 0.5 \times (1 + T_{AC}).$$

Solving the last two equations yields $T_{BC} = 3$ and $T_{AC} = 4$, leading to $T_{CC} = 5$.

- (c) Let

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_A & \mu_B & \mu_C \end{bmatrix}.$$

We have that

$$\frac{1}{T_{AA}} = 0.4, \quad \frac{1}{T_{BB}} = 0.4, \quad \frac{1}{T_{CC}} = 0.2,$$

which immediately implies that

$$\boldsymbol{\mu} = \begin{bmatrix} 0.4 & 0.4 & 0.2 \end{bmatrix}.$$

To show that μ is the invariant distribution for the chain, it suffices to show that $\boldsymbol{\mu P} = \boldsymbol{\mu}$. We have

$$\boldsymbol{\mu P} = \begin{bmatrix} 0.4 & 0.4 & 0.2 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0.5 & 0 & 0.5 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0.4 & 0.4 & 0.2 \end{bmatrix} = \boldsymbol{\mu},$$

and the conclusion follows.